Topology MATM36: Course Summary of Important Results

Simon Gustafsson

Metric Spaces

Definition 1. *Let X be a topological space and $A \subseteq X$. A point $x \in X$ is called an* adherent point *of A if every open neighbourhood U of x satisfies*

$$U \cap A \neq \varnothing.$$

In the special case where X is a metric space with metric d, this is equivalent to requiring that for every $r > 0$,

$$B_r(x) \cap A \neq \varnothing.$$

Definition 2. *Let X be a topological space and $A \subseteq X$. The set A is called* dense *in X if every nonempty open set $U \subseteq X$ satisfies*

$$U \cap A \neq \varnothing.$$

Equivalently, A is dense in X if $\overline{A} = X$.

Definition 3. *Let X be a topological space and $A \subseteq X$. The set A is called* nowhere dense *in X if*

$$\operatorname{int}(\overline{A}) = \varnothing.$$

Definition 4. *A topological space X is* separable *if: $\exists D \subseteq X : D$ countable and dense.*

Baire Category Theorem

Theorem 1 (Baire Category Theorem)**.** *Let (X, d) be a complete metric space and let $\{U_n\}_{n=1}^{\infty}$ be a sequence of dense open subsets of X. Then $\bigcap_{n=1}^{\infty} U_n$ is dense in X.*

Proof. Fix a nonempty open ball $B(x, \varepsilon) \subset X$.

Since U_1 is dense and open, choose $y_1 \in U_1 \cap B(x, \varepsilon)$ and $r_1 > 0$ such that

$$\overline{B}(y_1, r_1) \subset U_1 \cap B(x, \varepsilon), \qquad r_1 < 1.$$

Inductively, having chosen y_n, r_n, use that U_{n+1} is dense and open to pick $y_{n+1} \in U_{n+1} \cap B(y_n, r_n)$ and $r_{n+1} > 0$ such that

$$\overline{B}(y_{n+1}, r_{n+1}) \subset U_{n+1} \cap B(y_n, r_n), \qquad r_{n+1} < \frac{1}{n+1}.$$

Then the closed balls are nested:

$$\overline{B}(y_{n+1}, r_{n+1}) \subset \overline{B}(y_n, r_n) \subset \cdots \subset \overline{B}(y_1, r_1) \subset B(x, \varepsilon),$$

and for $m > n$ we have $y_m \in B(y_n, r_n)$, hence $d(y_m, y_n) < r_n \to 0$. Thus (y_n) is Cauchy, so by completeness $y_n \to y \in X$.

For each fixed n, all tails of the sequence lie in $\overline{B}(y_n, r_n)$, so the limit satisfies $y \in \overline{B}(y_n, r_n) \subset U_n$. Hence $y \in \bigcap_{n=1}^{\infty} U_n$, and also $y \in \overline{B}(y_1, r_1) \subset B(x, \varepsilon)$. Therefore

$$B(x, \varepsilon) \cap \bigcap_{n=1}^{\infty} U_n \neq \varnothing,$$

so the intersection is dense. □

Remark 1. *From the definitions it follows that a set $B \subseteq X$ satisfies $\operatorname{int}(B) = \varnothing \iff X \setminus B$ is dense in X. In particular, if A is nowhere dense, then $X \setminus \overline{A}$ is open and dense in X.*

Corollary 1. *Let (X, d) be a complete metric space and let $\{F_n\}_{n=1}^{\infty}$ be a sequence of nowhere dense subsets of X. Then*

$$\operatorname{int}\left(\bigcup_{n=1}^{\infty} F_n\right) = \varnothing.$$

Proof. Since F_n is nowhere dense, $U_n := X \setminus \overline{F_n}$ is open and dense; by Baire, $\bigcap_{n=1}^{\infty} U_n = \bigcup_{n=1}^{\infty} \overline{F_n}$ is dense, hence nonempty in every nonempty open set. □

Finite Products of Metric Spaces

Definition 5. *Let (X_k, d_k), $k = 1, \ldots, n$, be metric spaces and set*

$$X = X_1 \times \cdots \times X_n.$$

A metric d on X is said to be a product metric *if convergence in (X, d) is coordinatewise,*

$$x^{(m)} = (x_1^{(m)}, \ldots, x_n^{(m)}) \to x = (x_1, \ldots, x_n)$$

in X if and only if for each $k = 1, \ldots, n$,

$$x_k^{(m)} \to x_k \quad \text{in } X_k.$$

Theorem 2. *Suppose d is a metric on $X = X_1 \times \cdots \times X_n$ satisfying the above coordinatewise convergence property. Then the open sets in (X, d) are precisely the unions of product sets*

$$U_1 \times \cdots \times U_n,$$

where each U_k is open in X_k.

Proof. First, if $E_k \subset X_k$ is closed for each k, then

$$E_1 \times \cdots \times E_n$$

is closed in X. Indeed, if $x^{(m)} \in E_1 \times \cdots \times E_n$ and $x^{(m)} \to x$ in X, then by coordinatewise convergence, $x_k^{(m)} \to x_k$ in X_k. Since each E_k is closed, $x_k \in E_k$, hence $x \in E_1 \times \cdots \times E_n$.

Now let $U_k \subset X_k$ be open. Then

$$V^{(k)} := X_1 \times \cdots \times X_{k-1} \times U_k \times X_{k+1} \times \cdots \times X_n$$

is open in X, since its complement is of the form

$$X_1 \times \cdots \times (X_k \setminus U_k) \times \cdots \times X_n,$$

which is closed by the previous remark. Hence

$$U_1 \times \cdots \times U_n = \bigcap_{k=1}^{n} V^{(k)}$$

is open in X, and any union of such products is open.

Conversely, let $U \subset X$ be open and $x = (x_1, \ldots, x_n) \in U$. We show there exist open sets $U_k \subset X_k$ such that

$$x \in U_1 \times \cdots \times U_n \subset U.$$

Suppose not. Then for each $m \in \mathbb{N}$,

$$B_{X_1}(x_1, 1/m) \times \cdots \times B_{X_n}(x_n, 1/m) \not\subset U.$$

Thus there exists $x^{(m)} \notin U$ with

$$d_k(x_k^{(m)}, x_k) < \frac{1}{m}, \qquad k = 1, \ldots, n.$$

Then $x_k^{(m)} \to x_k$ for each k, hence $x^{(m)} \to x$ in X. Since $X \setminus U$ is closed and each $x^{(m)} \in X \setminus U$, we conclude $x \in X \setminus U$, a contradiction. Therefore such open sets U_k exist, and U is a union of product sets. □

Theorem 3 (Heine–Borel Theorem)**.** *For a subspace $E \subseteq \mathbb{R}^n$, the following are equivalent:*

(i) E is compact;

(ii) every sequence in E has a convergent subsequence with limit in E;

(iii) E is closed and bounded.

Proof. By the general compactness theorem for metric spaces, for any metric space the following are equivalent: compactness, sequential compactness, and completeness together with total boundedness.

Now let $E \subseteq \mathbb{R}^n$.

If E is closed and bounded, then E is complete (since a subspace of $\mathbb{R}^n$ is complete if and only if it is closed) and totally bounded (since a subset of $\mathbb{R}^n$ is totally bounded if and only if it is bounded). Hence E is compact, and therefore every sequence in E has a convergent subsequence.

Conversely, if E is compact, then by the same general theorem E is complete and totally bounded. Since E is a subspace of $\mathbb{R}^n$, completeness implies that E is closed, and total boundedness implies that E is bounded. Thus E is closed and bounded.

Therefore (i), (ii), and (iii) are equivalent. □

Theorem 4 (Uniform Continuity Theorem)**.** *Let (X, d) and (Y, ρ) be metric spaces, and suppose that X is compact. Then every continuous function $f : X \to Y$ is uniformly continuous.*

Proof. Suppose f is not uniformly continuous. Then there exists $\varepsilon > 0$ such that for every $m \in \mathbb{N}$ there are points $x_m, z_m \in X$ with

$$d(x_m, z_m) < \frac{1}{m} \qquad \text{but} \qquad \rho\big(f(x_m), f(z_m)\big) \geq \varepsilon.$$

Since X is compact, the sequence (x_m) has a convergent subsequence; passing to that subsequence, we may assume

$$x_m \to x \in X.$$

Because $d(x_m, z_m) \to 0$, also

$$z_m \to x.$$

By continuity of f at x,

$$f(x_m) \to f(x) \qquad \text{and} \qquad f(z_m) \to f(x).$$

Hence

$$\rho\big(f(x_m), f(z_m)\big) \leq \rho\big(f(x_m), f(x)\big) + \rho\big(f(x), f(z_m)\big) \to 0,$$

which contradicts

$$\rho\big(f(x_m), f(z_m)\big) \geq \varepsilon.$$

Therefore f is uniformly continuous. $\square$

Theorem 5 (Continuity and boundedness of linear operators)**.** *Let X and Y be normed spaces and let $T : X \to Y$ be linear. The following are equivalent:*

(i) T is continuous;

(ii) T is continuous at 0;

(iii) T is bounded, i.e. there exists $C > 0$ such that

$$\|T(x)\| \leq C\|x\| \quad \textit{for all } x \in X.$$

Proof. (i)⇒(ii) is immediate.

(ii)⇒(iii): Since T is continuous at 0, there exists $\delta > 0$ such that

$$\|x\| \leq \delta \;\Rightarrow\; \|T(x)\| \leq 1.$$

For arbitrary $x \neq 0$, set $y = \frac{\delta}{\|x\|}x$, so $\|y\| = \delta$ and hence

$$\|T(y)\| \leq 1.$$

By linearity,

$$\|T(x)\| = \frac{\|x\|}{\delta}\|T(y)\| \leq \frac{1}{\delta}\|x\|.$$

Thus T is bounded.

(iii)⇒(i): If $\|T(x)\| \leq C\|x\|$, then for all $x, x_0 \in X$,

$$\|T(x) - T(x_0)\| = \|T(x - x_0)\| \leq C\|x - x_0\|,$$

so T is Lipschitz, hence continuous. $\square$

Theorem 6 (Principle of Uniform Boundedness)**.** *Let X and Y be Banach spaces and let $\{T_\alpha\}_{\alpha\in A}$ be a family of continuous linear operators $T_\alpha : X \to Y$. Suppose that for each $x \in X$,*

$$\sup_{\alpha\in A} \|T_\alpha x\| < \infty.$$

Then

$$\sup_{\alpha\in A} \|T_\alpha\| < \infty.$$

Proof. For $n \in \mathbb{N}$, define

$$E_n := \left\{ x \in X : \sup_{\alpha\in A} \|T_\alpha x\| \leq n \right\}.$$

Then each E_n is closed, $E_n \subseteq E_{n+1}$, and by assumption

$$X = \bigcup_{n=1}^{\infty} E_n.$$

By the Baire Category Theorem, some E_m has nonempty interior. Thus there exist $x_0 \in X$ and $\varepsilon > 0$ such that

$$B(x_0, \varepsilon) \subseteq E_m.$$

Let $c := \sup_{\alpha\in A} \|T_\alpha x_0\| < \infty$. If $\|x\| < 1$, then $x_0 + \varepsilon x \in E_m$, hence

$$\|T_\alpha(x_0 + \varepsilon x)\| \leq m.$$

Thus

$$\varepsilon\|T_\alpha x\| = \|T_\alpha(x_0 + \varepsilon x) - T_\alpha(x_0)\| \leq m + c.$$

Therefore

$$\|T_\alpha x\| \leq \frac{m+c}{\varepsilon} \quad \text{for all } \|x\| < 1.$$

Hence $\|T_\alpha\| \leq \frac{m+c}{\varepsilon}$ for all $\alpha \in A$, and so

$$\sup_{\alpha\in A} \|T_\alpha\| < \infty.$$

□

Topologies

Definition 6. *Let X be a set. A* topology *on X is a collection $\mathcal{T} \subseteq \mathcal{P}(X)$ of subsets of X such that*

(i) $\varnothing \in \mathcal{T}$ *and* $X \in \mathcal{T}$,

(ii) if $\{U_\alpha\}_{\alpha\in I} \subseteq \mathcal{T}$, *then*

$$\bigcup_{\alpha\in I} U_\alpha \in \mathcal{T},$$

(iii) if $U_1, \dots, U_n \in \mathcal{T}$, *then*

$$\bigcap_{i=1}^{n} U_i \in \mathcal{T}.$$

The pair $(X, \mathcal{T})$ is called a topological space. *The sets $U \in \mathcal{T}$ are called* open sets. *A subset $F \subseteq X$ is called* closed *if its complement is open.*

Separation Axioms

Definition 7 (T_0 space). *A topological space $(X, \mathcal{T})$ is called a T_0-space if for every pair of distinct points $x, y \in X$, there exists an open set $U \in \mathcal{T}$ such that*

$$x \in U,\ y \notin U \quad \text{or} \quad y \in U,\ x \notin U.$$

Definition 8 (T_1 space). *A topological space $(X, \mathcal{T})$ is called a T_1-space if for every pair of distinct points $x, y \in X$, there exists an open set $U \in \mathcal{T}$ such that*

$$y \in U \quad \text{and} \quad x \notin U.$$

Equivalently, $(X, \mathcal{T})$ is T_1 if and only if every singleton set $\{x\}$ is closed in X.

Definition 9 (T_2 space / Hausdorff). *A topological space $(X, \mathcal{T})$ is called a T_2-space, or Hausdorff, if for every pair of distinct points $x, y \in X$, there exist open sets $U, V \in \mathcal{T}$ such that*

$$x \in U, \qquad y \in V, \qquad U \cap V = \varnothing.$$

Definition 10 (Regular space). *A topological space $(X, \mathcal{T})$ is called regular if for every closed set $F \subseteq X$ and every point $x \in X \setminus F$, there exist open sets $U, V \in \mathcal{T}$ such that*

$$x \in U, \qquad F \subseteq V, \qquad U \cap V = \varnothing.$$

Definition 11 (T_3 space). *A topological space $(X, \mathcal{T})$ is called a T_3-space if it is both T_1 and regular.*

Definition 12 (Normal space). *A topological space $(X, \mathcal{T})$ is called normal if for any two disjoint closed sets $F, G \subseteq X$, there exist open sets $U, V \in \mathcal{T}$ such that*

$$F \subseteq U, \qquad G \subseteq V, \qquad U \cap V = \varnothing.$$

Definition 13 (T_4 space). *A topological space $(X, \mathcal{T})$ is called a T_4-space if it is both T_1 and normal.*

Theorem 7. *Every compact Hausdorff space is normal. In particular, every compact Hausdorff space is a T_4-space.*

Proof. Let $F, G \subseteq X$ be disjoint closed sets. Since X is Hausdorff, for each $x \in F$ and $y \in G$ there exist disjoint open sets $U_{x,y}$ and $V_{x,y}$ such that

$$x \in U_{x,y}, \qquad y \in V_{x,y}, \qquad U_{x,y} \cap V_{x,y} = \varnothing.$$

For fixed $x \in F$, the sets $\{V_{x,y}\}_{y \in G}$ form an open cover of G. Since G is closed in the compact space X, it is compact, so we may choose a finite subcover

$$G \subseteq V_{x,y_1} \cup \cdots \cup V_{x,y_n}.$$

Let

$$U_x := U_{x,y_1} \cap \cdots \cap U_{x,y_n}, \qquad V_x := V_{x,y_1} \cup \cdots \cup V_{x,y_n}.$$

Then $x \in U_x$, $G \subseteq V_x$, and $U_x \cap V_x = \varnothing$.

The family $\{U_x\}_{x \in F}$ is an open cover of F. Since F is compact, choose a finite subcover

$$F \subseteq U_{x_1} \cup \cdots \cup U_{x_m}.$$

Define

$$U := U_{x_1} \cup \cdots \cup U_{x_m}, \qquad V := V_{x_1} \cap \cdots \cap V_{x_m}.$$

Then U and V are open, $F \subseteq U$, $G \subseteq V$, and $U \cap V = \varnothing$. Hence X is normal. $\square$

Remark 2 (Continuity criterion in metric spaces). *Let $(X, \mathcal{T})$ be a topological space and (Y, d) a metric space. A function $f : X \to Y$ is continuous at $x \in X$ if and only if for every $\varepsilon > 0$ there exists a neighbourhood U of x such that*

$$d(f(y), f(x)) < \varepsilon \quad \textit{for all } y \in U.$$

Equivalently, $f(U) \subseteq B(f(x), \varepsilon)$.

Theorem 8 (Uniform limit preserves continuity). *Let $(X, \mathcal{T})$ be a topological space and (Y, d) a metric space. Suppose (f_n) is a sequence of functions $f_n : X \to Y$ converging uniformly to $f : X \to Y$. If each f_n is continuous at $x \in X$, then f is continuous at x.*

Proof. We first recall the following characterization of continuity at a point: f is continuous at x if and only if for every $\varepsilon > 0$ there exists a neighbourhood U of x such that

$$d(f(y), f(x)) < \varepsilon \quad \text{for all } y \in U.$$

Now let $\varepsilon > 0$. By uniform convergence, there exists $N \in \mathbb{N}$ such that

$$d(f_N(y), f(y)) < \frac{\varepsilon}{3} \quad \text{for all } y \in X.$$

Since f_N is continuous at x, there exists a neighbourhood U of x such that

$$d(f_N(y), f_N(x)) < \frac{\varepsilon}{3} \quad \text{for all } y \in U.$$

Then for all $y \in U$,

$$d(f(y), f(x)) \leq d(f(y), f_N(y)) + d(f_N(y), f_N(x)) + d(f_N(x), f(x)) < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.$$

By the above characterization, f is continuous at x. □

Theorem 9. *Every metric space (X, d) is a T_4-space.*

Proof. First, for each $x \in X$,

$$\{x\} = \bigcap_{r>0} \overline{B(x, r)},$$

so singletons are closed and X is T_1.

Let $E, F \subseteq X$ be disjoint closed sets. For each $x \in E$, there exists $r(x) > 0$ such that

$$B(x, r(x)) \cap F = \varnothing.$$

Similarly, for each $y \in F$, there exists $r(y) > 0$ such that

$$B(y, r(y)) \cap E = \varnothing.$$

Define

$$U := \bigcup_{x \in E} B\Big(x, \tfrac{r(x)}{2}\Big), \quad V := \bigcup_{y \in F} B\Big(y, \tfrac{r(y)}{2}\Big).$$

Then U, V are open, $E \subseteq U$, $F \subseteq V$.

Suppose $z \in U \cap V$. Then there exist $x \in E$, $y \in F$ such that

$$d(x, z) < \tfrac{r(x)}{2}, \quad d(y, z) < \tfrac{r(y)}{2}.$$

Hence

$$d(x, y) \leq d(x, z) + d(z, y) < \tfrac{r(x)+r(y)}{2} \leq \max\{r(x), r(y)\}.$$

Thus either $d(x, y) < r(x)$ or $d(x, y) < r(y)$, contradicting the choice of $r(x)$ or $r(y)$. Therefore $U \cap V = \varnothing$, and X is normal. □

Theorem 10 (Urysohn's Lemma). *Let E and F be disjoint closed subsets of a normal space X. Then there exists a continuous function*

$$f : X \to [0,1]$$

such that

$$f|_E = 0, \qquad f|_F = 1.$$

Proof. Set $V := X \setminus F$, which is open and contains E. Using normality repeatedly, construct for each dyadic rational $r \in (0,1)$ an open set U_r such that

$$E \subseteq U_r \subseteq V$$

and

$$\overline{U_r} \subseteq U_s \qquad \text{whenever } 0 < r < s < 1.$$

Define $f : X \to [0,1]$ by

$$f(x) = \begin{cases} 0, & x \in \bigcap_{0<r<1} U_r, \\ \sup\{r \in (0,1) \cap D : x \notin U_r\}, & \text{otherwise}, \end{cases}$$

where D denotes the set of dyadic rationals in $(0,1)$.

Then $0 \le f \le 1$, and since $E \subseteq U_r$ for every r, we have $f = 0$ on E. Also $U_r \subseteq X \setminus F$ for every r, so $f = 1$ on F.

To prove continuity, let $x \in X$ and $\varepsilon > 0$. Choose dyadic rationals r, s such that

$$f(x) - \varepsilon < r < f(x) < s < f(x) + \varepsilon.$$

Then $x \notin \overline{U_r}$ and $x \in U_s$. Hence

$$W := U_s \setminus \overline{U_r}$$

is an open neighbourhood of x. For every $y \in W$ we have $y \notin U_r$, so $f(y) \ge r$, and $y \in U_s$, so $f(y) \le s$. Thus

$$r \le f(y) \le s,$$

and therefore

$$|f(y) - f(x)| < \varepsilon.$$

So f is continuous. □

Theorem 11 (Tietze Extension Theorem). *Let X be a normal topological space, $Y \subseteq X$ closed, and let $f : Y \to \mathbb{R}$ be bounded and continuous. Then there exists a bounded continuous function $h : X \to \mathbb{R}$ such that $h|_Y = f$.*

Proof. Assume $|f| \le c_0$ on Y. Using Urysohn's Lemma, construct a continuous function $g_0 : X \to \mathbb{R}$ such that

$$|g_0| \le \tfrac{c_0}{3} \quad \text{on } X, \qquad |f - g_0| \le \tfrac{2c_0}{3} \quad \text{on } Y.$$

Inductively, having defined $g_0, \ldots, g_{n-1}$, set

$$c_n := \sup_{y \in Y} |f(y) - g_0(y) - \cdots - g_{n-1}(y)|.$$

Again by Urysohn, construct g_n such that

$$|g_n| \leq \tfrac{c_n}{3} \quad \text{on } X, \qquad |f - g_0 - \cdots - g_n| \leq \tfrac{2c_n}{3} \quad \text{on } Y.$$

Then $c_n \leq (2/3)^n c_0$ and $|g_n| \leq (2/3)^n c_0/3$.

The series $\sum g_n$ converges uniformly on X to a continuous function

$$h := \sum_{n=0}^{\infty} g_n.$$

Moreover, on Y,

$$f = \lim_{n\to\infty} (g_0 + \cdots + g_n) = h,$$

and h is bounded. Thus h is the desired extension. □

Basis & Subbasis

Definition 14. *Let X be a set. A collection $\mathcal{B}$ of subsets of X is called a* basis *for a topology on X if:*

(i) For every $x \in X$ there exists $B \in \mathcal{B}$ such that $x \in B$.

(ii) If $x \in B_1 \cap B_2$ for $B_1, B_2 \in \mathcal{B}$, then there exists $B_3 \in \mathcal{B}$ such that

$$x \in B_3 \subseteq B_1 \cap B_2.$$

Theorem 12 (Topology generated by a basis)**.** *If $\mathcal{B}$ satisfies the two conditions above, then there exists a unique topology $\mathcal{T}$ on X whose open sets are precisely the unions of elements of $\mathcal{B}$. That is,*

$$\mathcal{T} = \left\{ \bigcup_{\alpha\in I} B_\alpha : B_\alpha \in \mathcal{B},\ I \text{ arbitrary} \right\}.$$

Definition 15. *Let X be a set. A collection $\mathcal{S}$ of subsets of X is called a* subbasis *for a topology on X if the union of all sets in $\mathcal{S}$ equals X,*

$$X = \bigcup_{S\in\mathcal{S}} S.$$

The topology generated by $\mathcal{S}$ is obtained by first forming the basis consisting of all finite intersections of elements of $\mathcal{S}$,

$$\mathcal{B} = \{S_1 \cap \cdots \cap S_n : S_i \in \mathcal{S},\ n \in \mathbb{N}\},$$

and then taking arbitrary unions of these sets.

The topology generated by the subbasis $\mathcal{S}$ consists of all arbitrary unions of elements of this basis. That is, a set $U \subseteq X$ is open if and only if it can be written as

$$U = \bigcup_{\alpha\in I} \left(\bigcap_{i=1}^{n_\alpha} S_{\alpha,i} \right), \qquad S_{\alpha,i} \in \mathcal{S}, \quad n_\alpha \in \mathbb{N}.$$

Theorem 13 (Basis criterion). *A collection $\mathcal{B}$ of subsets of X is a basis for a topology on X if and only if:*

(i) For every $x \in X$, there exists $B \in \mathcal{B}$ such that $x \in B$;

(ii) If $B_1, B_2 \in \mathcal{B}$ and $x \in B_1 \cap B_2$, then there exists $B_3 \in \mathcal{B}$ such that

$$x \in B_3 \subseteq B_1 \cap B_2.$$

Proof. ($\Rightarrow$) If $\mathcal{B}$ is a basis, then (i) holds since X is open, and (ii) holds since $B_1 \cap B_2$ is open and can be written as a union of basis elements.

($\Leftarrow$) Suppose (i) and (ii) hold. Let $\mathcal{T}$ be the collection of all unions of sets in $\mathcal{B}$. Then $\varnothing, X \in \mathcal{T}$ and arbitrary unions are in $\mathcal{T}$.

To show closure under finite intersections, let $U, V \in \mathcal{T}$ and $x \in U \cap V$. Then there exist $B_1, B_2 \in \mathcal{B}$ such that

$$x \in B_1 \subseteq U, \quad x \in B_2 \subseteq V.$$

By (ii), there exists $B_3 \in \mathcal{B}$ such that

$$x \in B_3 \subseteq B_1 \cap B_2 \subseteq U \cap V.$$

Hence

$$U \cap V = \bigcup_{x \in U \cap V} B_3(x),$$

so $U \cap V \in \mathcal{T}$. Thus $\mathcal{T}$ is a topology and $\mathcal{B}$ is a basis. □

Theorem 14 (Lindelöf's Theorem). *Let X be a second-countable topological space. Then every open cover of X admits a countable subcover.*

Proof. Let $\mathcal{B} = \{B_n\}_{n \in \mathbb{N}}$ be a countable basis for X, and let $\mathcal{U} = \{U_\alpha\}_{\alpha \in A}$ be an open cover of X.

For each $n \in \mathbb{N}$, since $B_n \subseteq X = \bigcup_{\alpha \in A} U_\alpha$, we may choose $\alpha(n) \in A$ such that

$$B_n \subseteq U_{\alpha(n)}.$$

Then the collection $\{U_{\alpha(n)}\}_{n \in \mathbb{N}}$ is countable.

To see that it covers X, let $x \in X$. Since $\mathcal{B}$ is a basis, there exists n such that $x \in B_n$. By construction, $B_n \subseteq U_{\alpha(n)}$, so $x \in U_{\alpha(n)}$. Hence

$$X = \bigcup_{n=1}^{\infty} U_{\alpha(n)}.$$

Thus $\{U_{\alpha(n)}\}_{n \in \mathbb{N}}$ is a countable subcover. □

Compactness

Theorem 15. *Let S be a compact subset of a Hausdorff space X. For each $x \in X \setminus S$, there exist disjoint open sets $U, V \subseteq X$ such that*

$$x \in U, \qquad S \subseteq V.$$

Proof. For each $y \in S$, since X is Hausdorff, there exist disjoint open sets U_y and V_y such that
$$x \in U_y, \qquad y \in V_y.$$
Then $\{V_y\}_{y \in S}$ is an open cover of S. By compactness, there exist $y_1, \ldots, y_n \in S$ such that
$$S \subseteq V_{y_1} \cup \cdots \cup V_{y_n} =: V.$$
Let
$$U := U_{y_1} \cap \cdots \cap U_{y_n}.$$
Then U is open, $x \in U$, and $U \cap V = \varnothing$. □

Corollary 2. *A compact subset of a Hausdorff space is closed.*

Proof. Let $S \subseteq X$ be compact and $x \in X \setminus S$. By the lemma, there exists an open set U containing x with $U \cap S = \varnothing$. Hence $X \setminus S$ is open, so S is closed. □

Theorem 16 (Theorem 6.5)**.** *Every compact Hausdorff space is normal.*

Proof. Let $S, T \subseteq X$ be disjoint closed sets. Since X is compact, S and T are compact. For each $x \in T$, by the lemma on separating a point from a compact set, there exist disjoint open sets U_x, V_x such that
$$x \in U_x, \qquad S \subseteq V_x.$$
The family $\{U_x\}_{x \in T}$ is an open cover of T, so by compactness there exist $x_1, \ldots, x_n \in T$ such that
$$T \subseteq U_{x_1} \cup \cdots \cup U_{x_n} =: U.$$
Set
$$V := V_{x_1} \cap \cdots \cap V_{x_n}.$$
Then U and V are open, $T \subseteq U$, $S \subseteq V$, and $U \cap V = \varnothing$. Hence X is normal. □

Theorem 17 (Theorem 6.6)**.** *Let $f : X \to Y$ be continuous, where X is compact and Y is a topological space. Then $f(X)$ is compact in Y.*

Proof. Let $\{U_\alpha\}_{\alpha \in A}$ be an open cover of $f(X)$. Then
$$X = \bigcup_{\alpha \in A} f^{-1}(U_\alpha),$$
and each $f^{-1}(U_\alpha)$ is open in X by continuity of f. Since X is compact, there exist $\alpha_1, \ldots, \alpha_n \in A$ such that
$$X = f^{-1}(U_{\alpha_1}) \cup \cdots \cup f^{-1}(U_{\alpha_n}).$$
Applying f, we get
$$f(X) \subseteq U_{\alpha_1} \cup \cdots \cup U_{\alpha_n}.$$
Thus $\{U_{\alpha_1}, \ldots, U_{\alpha_n}\}$ is a finite subcover of $f(X)$, so $f(X)$ is compact. □

Theorem 18 (Theorem 6.7)**.** *Let $f : X \to Y$ be a continuous map from a compact space X to a Hausdorff space Y. If f is one-to-one, then f is a homeomorphism of X onto $f(X)$.*

Proof. Since f is continuous and bijective onto $f(X)$, it suffices to show that $f^{-1} : f(X) \to X$ is continuous, or equivalently that $f(U)$ is open in $f(X)$ whenever U is open in X.

Let $U \subseteq X$ be open. Then $X \setminus U$ is closed in X, hence compact. By continuity, $f(X \setminus U)$ is compact in Y, and since Y is Hausdorff, it is closed in Y. Therefore it is closed in $f(X)$.

But

$$f(X \setminus U) = f(X) \setminus f(U)$$

since f is injective. Hence $f(U)$ is open in $f(X)$, and f^{-1} is continuous. Thus f is a homeomorphism onto $f(X)$. □

Arzelà–Ascoli Theorem

Lemma 1. *Let X be compact, and let $(f_n) \subset C(X)$ be uniformly bounded and equicontinuous. Then there exists a subsequence (f_{n_k}) which converges uniformly on X.*

Proof. For each $m \in \mathbb{N}$ and each $x \in X$, equicontinuity gives an open neighbourhood $U_{x,m}$ such that

$$|f_n(y) - f_n(x)| < \frac{1}{m} \qquad \text{for all } y \in U_{x,m},\ n \in \mathbb{N}.$$

By compactness, for each m there exist points $x_{m,1}, \dots, x_{m,r_m}$ such that

$$X = \bigcup_{i=1}^{r_m} U_{x_{m,i},m}.$$

Let

$$A := \bigcup_{m=1}^{\infty} \{x_{m,1}, \dots, x_{m,r_m}\}.$$

Since this is a countable union of finite sets, we may write

$$A = \{a_1, a_2, a_3, \dots\}.$$

Since (f_n) is uniformly bounded, for each j the sequence $(f_n(a_j))$ is bounded, hence admits a convergent subsequence. We now construct nested subsequences

$$(f_n) \supset (f_n^{(1)}) \supset (f_n^{(2)}) \supset (f_n^{(3)}) \supset \cdots$$

such that $(f_n^{(k)}(a_k))$ converges for each k.

$$\begin{aligned}
(f_n^{(1)}) &= (f_{n_1}^{(1)}, f_{n_2}^{(1)}, f_{n_3}^{(1)}, \dots),\\
(f_n^{(2)}) &= (f_{n_1}^{(2)}, f_{n_2}^{(2)}, f_{n_3}^{(2)}, \dots),\\
(f_n^{(3)}) &= (f_{n_1}^{(3)}, f_{n_2}^{(3)}, f_{n_3}^{(3)}, \dots),\\
&\vdots
\end{aligned}$$

where each row is a subsequence of the previous one.

Define the diagonal subsequence by $g_k := f_{n_k^{(k)}}$. Fix j. For all $k \geq j$, the term g_k lies in the subsequence $(f_n^{(j)})$, hence $(g_k(a_j))$ converges. Thus $(g_k(a_j))$ converges for every $a_j \in A$.

We now show that (g_k) is uniformly Cauchy. Let $\varepsilon > 0$ and choose m such that $\frac{3}{m} < \varepsilon$. From the finite cover

$$X = \bigcup_{i=1}^{r_m} U_{x_{m,i},m},$$

and since each $x_{m,i} \in A$, the sequences $(g_k(x_{m,i}))$ converge. Hence there exists N such that for all $k, \ell \geq N$,

$$|g_k(x_{m,i}) - g_\ell(x_{m,i})| < \frac{1}{m} \qquad \text{for } i = 1, \dots, r_m.$$

Now let $x \in X$. Choose i such that $x \in U_{x_{m,i},m}$. Then

$$\begin{aligned} |g_k(x) - g_\ell(x)| &\leq |g_k(x) - g_k(x_{m,i})| + |g_k(x_{m,i}) - g_\ell(x_{m,i})| + |g_\ell(x_{m,i}) - g_\ell(x)| \\ &< \frac{1}{m} + \frac{1}{m} + \frac{1}{m} = \frac{3}{m} \\ &< \varepsilon \end{aligned}$$

Since x arbitrary (g_k) is uniformly Cauchy, hence converges uniformly on X. □

Theorem 19 (Arzelà–Ascoli)**.** *Let X be a compact topological space and let $\mathcal{F} \subset C(X)$. Then $\overline{\mathcal{F}}$ is compact in $(C(X), \|\cdot\|_\infty) \Leftrightarrow \mathcal{F}$ is uniformly bounded and equicontinuous.*

Proof. ($\Leftarrow$) Assume $\mathcal{F}$ is uniformly bounded and equicontinuous. Then by the lemma, every sequence in $\mathcal{F}$ has a uniformly convergent subsequence, hence $\overline{\mathcal{F}}$ sequentially compact.

($\Rightarrow$) Assume $\mathcal{F}$ is precompact. Then $\mathcal{F}$ is totally bounded.

Uniform boundedness: For $\varepsilon = 1$, there exist $f_1, \dots, f_n$ such that

$$\mathcal{F} \subset \bigcup_{k=1}^{n} B(f_k, 1).$$

Hence $\|f\|_\infty \leq 1 + \max_k \|f_k\|_\infty$ for all $f \in \mathcal{F}$.

Equicontinuity: Let $\varepsilon > 0$ and choose $f_1, \dots, f_n$ such that

$$\mathcal{F} \subset \bigcup_{k=1}^{n} B(f_k, \varepsilon).$$

Fix $x \in X$. Since each f_k is continuous, there exists a neighbourhood U of x such that

$$|f_k(y) - f_k(x)| < \varepsilon \quad \forall y \in U,\ k = 1, \dots, n.$$

If $f \in \mathcal{F}$, choose k with $\|f - f_k\|_\infty < \varepsilon$. Then

$$|f(y) - f(x)| \leq |f(y) - f_k(y)| + |f_k(y) - f_k(x)| + |f_k(x) - f(x)| < \varepsilon + \varepsilon + \varepsilon = 3\varepsilon.$$

□

Locally Compact

Theorem 20 (One-point compactification)**.** *Let X be a locally compact Hausdorff space. Then there exists a space $Y = X \cup \{\infty\}$ and a topology on Y such that:*

- *Y is compact and Hausdorff,*
- *the subspace topology on X coincides with the original topology,*
- *the topology is uniquely determined.*

The open sets in Y are:

- *all open sets of X,*
- *all sets of the form $U \cup \{\infty\}$ where $X \setminus U$ is compact in X.*

Connected

Theorem 21 (Theorem 8.1)**.** *Let f be a continuous function from a connected topological space X to a topological space Y. Then $f(X)$ is connected.*

Proof. Let $E \subseteq f(X)$ be both open and closed in $f(X)$. Then $f^{-1}(E)$ is both open and closed in X since f is continuous.

Because X is connected, we must have either $f^{-1}(E) = \varnothing$ or $f^{-1}(E) = X$.

In the first case, $E = \varnothing$. In the second case, $E = f(X)$. Hence $f(X)$ is connected. □

Theorem 22 (Theorem 8.2)**.** *Let $\{E_\alpha\}_{\alpha \in A}$ be a family of connected subsets of a topological space X such that $E_\alpha \cap E_\beta \neq \varnothing$ for all α, β. Then $\bigcup_{\alpha \in A} E_\alpha$ is connected.*

Proof. Let $E = \bigcup_\alpha E_\alpha$ and let $F \subseteq E$ be clopen with $F \neq \varnothing$. Pick $x \in F \cap E_{\alpha_0}$. Since E_{α_0} is connected, $F \cap E_{\alpha_0} = E_{\alpha_0}$, so $E_{\alpha_0} \subseteq F$.

For any β, $F \cap E_\beta$ is clopen in E_β and contains $E_{\alpha_0} \cap E_\beta \neq \varnothing$, hence $E_\beta \subseteq F$. Thus $F = E$. □

Definition 16. *Let X be a topological space and $x \in X$. The* connected component *of x, denoted $C(x)$, is the union of all connected subsets of X that contain x.*

Theorem 23. *$C(x)$ is connected and is the largest connected subset of X containing x.*

Theorem 24 (Theorem 8.3)**.** *Two connected components of X either coincide or are disjoint. Hence, the connected components form a partition of X.*

Theorem 25 (Theorem 8.4)**.** *Any interval in $\mathbb{R}$ (closed, open, or half-open; finite or infinite) is connected.*

Proof. It suffices to show that $[0, 1]$ is connected. Let $E \subseteq [0, 1]$ be clopen and assume $E \neq \varnothing$. Set

$$t = \sup E.$$

Since E is closed, $t \in E$. If $t < 1$, then since E is open, there exists $\varepsilon > 0$ such that $(t, t + \varepsilon) \subseteq E$, contradicting the definition of t. Hence $E = [0, 1]$. □

Path Connectedness

Theorem 26 (Lemma 9.1)**.** *The relation "there is a path in X from x to y" is an equivalence relation on X.*

Proof. We verify the three properties:

(i) **Reflexive:** For any $x \in X$, the constant map

$$\alpha : [0,1] \to X, \qquad \alpha(s) = x$$

is a path from x to x.

(ii) **Symmetric:** If α is a path from x to y, then

$$\beta : [0,1] \to X, \qquad \beta(s) = \alpha(1-s)$$

is a path from y to x.

(iii) **Transitive:** If α is a path from x to y and β is a path from y to z, define

$$\gamma(s) = \begin{cases} \alpha(2s), & 0 \le s \le \frac{1}{2}, \\ \beta(2s-1), & \frac{1}{2} \le s \le 1. \end{cases}$$

Then γ is a path from x to z.

Hence the relation is an equivalence relation. □

Theorem 27 (Theorem 9.2)**.** *Every path-connected space is connected.*

Proof. Fix $x_0 \in X$. For each $x \in X$, let $\gamma_x : [0,1] \to X$ be a path from x_0 to x. Since $[0,1]$ is connected and γ_x is continuous, $\gamma_x([0,1])$ is connected. Also, each $\gamma_x([0,1])$ contains x_0. Hence

$$X = \bigcup_{x \in X} \gamma_x([0,1])$$

is a union of connected subsets with a common point, so X is connected. □

Corollary 3 (Corollory 9.3)**.** *Each connected component of a topological space is a union of path components.*

Proof. A path component is path-connected, hence connected by the theorem. Therefore each path component is contained in some connected component. It follows that a connected component is the union of all path components contained in it. □

Finite Product Spaces

Theorem 28 (10.1)**.** *Let X be the product of the topological spaces $X_1, \dots, X_n$, and let π_j be the projection of X onto X_j. The product topology for X is the smallest topology for which each of the projections π_j is continuous.*

Proof. Let $\mathcal{T}$ be another topology for X such that the projections π_j are $\mathcal{T}$-continuous. Let $U_1, \ldots, U_n$ be open subsets of $X_1, \ldots, X_n$, respectively. Then each $\pi_j^{-1}(U_j)$ belongs to $\mathcal{T}$. Since

$$\pi_1^{-1}(U_1) \cap \cdots \cap \pi_n^{-1}(U_n) = U_1 \times \cdots \times U_n,$$

the basic set $U_1 \times \cdots \times U_n$ belongs to $\mathcal{T}$ and $\mathcal{T}$ includes the product topology. $\square$

Theorem 29 (10.2)**.** *Let $X = X_1 \times \cdots \times X_n$ be the product of the topological spaces $X_1, \ldots, X_n$. Then each projection π_j is an open map of X onto X_j.*

Proof. If $U = U_1 \times \cdots \times U_n$ is a nonempty basic open set, then $\pi_j(U) = U_j$ is open. Since maps preserve unions, the image of any open set is open. $\square$

Theorem 30 (10.3)**.** *Let E be a topological space and let f be a function from E to the product*

$$X = X_1 \times \cdots \times X_n.$$

Then f is continuous if and only if each $\pi_j \circ f$ is continuous, $1 \leq j \leq n$.

Proof. If f is continuous, then $\pi_j \circ f$ is also continuous since the composition of continuous functions is continuous.

Conversely, suppose that each $\pi_j \circ f$ is continuous. Let

$$U = U_1 \times \cdots \times U_n$$

be a basic open set in X. Then

$$f^{-1}(U) = (\pi_1 \circ f)^{-1}(U_1) \cap \cdots \cap (\pi_n \circ f)^{-1}(U_n)$$

is a finite intersection of open sets and hence is open. Since inverses of functions preserve unions, the inverse image of any open set is open. Hence f is continuous. $\square$

Theorem 31 (10.4)**.** *If $X_1, \ldots, X_n$ are Hausdorff spaces, then $X = X_1 \times \cdots \times X_n$ is Hausdorff.*

Proof. Let $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_n)$ be distinct points of X. Choose an index j such that $x_j \neq y_j$. Let U_j and V_j be disjoint open neighborhoods of x_j and y_j, respectively, in X_j. Then $\pi_j^{-1}(U_j)$ and $\pi_j^{-1}(V_j)$ are disjoint open neighborhoods of x and y, respectively, in X. $\square$

Theorem 32 (10.5)**.** *If X_j is a path-connected topological space, $1 \leq j \leq n$, then $X = X_1 \times \cdots \times X_n$ is path-connected.*

Proof. Let $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_n)$ be points of X. For $1 \leq j \leq n$, let

$$\gamma_j : [0, 1] \to X_j$$

be a path from x_j to y_j. Then the path

$$\gamma : [0, 1] \to X, \qquad \gamma(t) = (\gamma_1(t), \ldots, \gamma_n(t)), \qquad 0 \leq t \leq 1,$$

is a path from x to y. Hence X is path-connected. $\square$

Theorem 33 (10.6)**.** *If X_j is a connected topological space, $1 \leq j \leq n$, then $X = X_1 \times \cdots \times X_n$ is also connected.*

Proof. Let E be a closed and open subset of X. Replacing E by its complement, if necessary, we can assume that $E \neq \varnothing$. We must show that $E = X$.

Let $z = (z_1, \ldots, z_n) \in E$ and let $y = (y_1, \ldots, y_n) \in X$. The intersection

$$E \cap (X_1 \times \{z_2\} \times \cdots \times \{z_n\})$$

is a relatively open and closed subset of the coordinate slice, which includes z. Since X_1 is connected, so is the coordinate slice. It follows that E must include the entire coordinate slice. In particular,

$$(y_1, z_2, \ldots, z_n) \in E.$$

Now we repeat this argument in the second coordinate and continue until eventually we conclude that $y \in E$. Since $y \in X$ is arbitrary, we obtain $E = X$. □

Lemma 2 (10.7)**.** *Let Y be a topological space and let $\mathcal{B}$ be a base for the topology of Y. If every open cover of Y by sets in $\mathcal{B}$ has a finite subcover, then Y is compact.*

Proof. Let $\{U_\beta\}_{\beta \in B}$ be an open cover of Y. For each $y \in Y$, choose $V_y \in \mathcal{B}$ and an index β such that

$$y \in V_y \subset U_\beta.$$

The family $\{V_y : y \in Y\}$ forms an open cover of Y by sets in $\mathcal{B}$. By hypothesis, there exist a finite number of the V_y's that cover Y. Since each of these V_y's is contained in at least one of the U_β's, we obtain a finite number of the U_β's that cover Y. Hence Y is compact. □

Theorem 34 (10.8, Tychonoff's Theorem, finite case)**.** *If $X_1, \ldots, X_n$ are compact spaces, then $X_1 \times \cdots \times X_n$ is compact.*

Proof. It suffices to prove the theorem in the case where $n = 2$, and so we consider only

$$X = X_1 \times X_2.$$

Let $\mathcal{C}$ be a cover of $X_1 \times X_2$ by rectangular open sets of the form $U \times V$, where U is open in X_1 and V is open in X_2. By Lemma 10.7, it suffices to show that the cover $\mathcal{C}$ has a finite subcover.

Fix $z \in X_2$. The coordinate slice $X_1 \times \{z\}$ is compact. Hence there are a finite number of sets

$$U_1 \times V_1, \ldots, U_m \times V_m \in \mathcal{C}$$

that cover $X_1 \times \{z\}$. We can assume that $z \in V_j$ for all j, $1 \leq j \leq m$. Then

$$V(z) = V_1 \cap \cdots \cap V_m$$

is an open neighborhood of z in X_2. Furthermore,

$$\pi_2^{-1}(V(z))$$

is covered by a finite number of sets in $\mathcal{C}$, namely the sets

$$U_j \times V_j, \qquad 1 \leq j \leq m.$$

The sets $\{V(z) : z \in X_2\}$ form an open cover of X_2. Since X_2 is compact, there exist

$$z_1, \ldots, z_k \in X_2$$

such that

$$X_2 = V(z_1) \cup \cdots \cup V(z_k).$$

Then

$$X = \pi_2^{-1}(V(z_1)) \cup \cdots \cup \pi_2^{-1}(V(z_k)).$$

Since each $\pi_2^{-1}(V(z_j))$ can be covered by a finite number of sets in $\mathcal{C}$, and since the aggregate of these sets covers X, $\mathcal{C}$ has a finite subcover. $\square$